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Digital rights protected content playing

Abstract

A system for performing digital rights protected content playing includes a computing platform having a hardware processor and a memory storing a software code. The hardware processor executes the software code to receive, from a first user, metadata identifying one or more content segment(s), determine whether the first user has a right to access the content segment(s), produce a playlist using the metadata when the first user has the right, and output the playlist to the first user and/or a second user. The hardware processor may further execute the software code to receive the playlist from the second user, determine each content segment identified by the playlist that the second user has a right to access, generate a content compilation that omits any content segment identified by the playlist to which the second user lacks the right, and playout the content compilation to the second user.

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Background/Summary

(1) The present application is a Continuation of U.S. application Ser. No. 16/882,330, filed May 22, 2020.

BACKGROUND

(1) Media content in the form of movie content and television (TV) programming content, for example, is widely sought out and enjoyed by consumers. One source of pleasure for many consumers of movie and TV programming content is the ability to quickly access a favorite movie or TV scene without playing the movie or TV program that includes the scene in its entirety. Another source of pleasure for many consumers is playing back favorite scenes with friends or family.

(2) The desire by a consumer to gain quick play of preferred portions of content for personal enjoyment, may cause the consumer to make a personalized content clip by denoting favorite segments of content from the original and stringing those segments together. That personalized content clip can then be sent to friends or family, or posted to a social media platform. Unfortunately, those often innocently intended practices stray into illegality when digital rights protected content is copied and distributed to consumers who lack a right to access that content, such as a license or other entitlement to the content. Although legally playable content clips can be obtained from some sources, those clips are typically produced with promotional intent and reflect the editorial preferences of the owner or owners of the content, rather than its consumers. Consequently, there is a need in the art for a solution enabling the legal playing of personalized content clips produced from digital rights protected content.

SUMMARY

(3) There are provided systems and methods for performing digital rights protected content playing, substantially as shown in and/or described in connection with at least one of the figures, and as set forth more completely in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows an exemplary system for performing digital rights protected content playing, according to one implementation;
- (2) FIG. 2 shows a more detailed exemplary representation of a software code suitable for use in the system of FIG. 1, according to one implementation;
- (3) FIG. 3 shows an exemplary content editing pane provided by a graphical user interface (GUI) of the system in FIG. 1, according to one implementation;
- (4) FIG. 4 shows an exemplary introductory message pane provided by the GUI of the system in FIG. 1, according to one implementation;
- (5) FIG. 5A shows a flowchart presenting an exemplary method for performance by the system in FIG. 1, according to one implementation;
- (6) FIG. 5B shows a flowchart presenting exemplary actions for performing digital rights protected content playing, according to one implementation; and
- (7) FIG. 6 shows an exemplary content promotion pane provided by the GUI of the system in FIG. 1, according to one implementation.

DETAILED DESCRIPTION

(8) The following description contains specific information pertaining to implementations in the present disclosure. One skilled in the art will recognize that the present disclosure may be implemented in a manner different from that specifically discussed herein. The drawings in the present application and their accompanying detailed description are directed to merely exemplary implementations. Unless noted otherwise, like or corresponding elements among the figures may be indicated by like or corresponding reference numerals. Moreover, the drawings and illustrations in the present application are generally not to scale, and are not intended to correspond to actual relative dimensions.

(9) The present application discloses systems and methods for performing digital rights protected content playing that address and overcome the deficiencies in the conventional art. As stated above, the desire by a consumer to gain quick access to preferred portions of content for personal enjoyment may cause the consumer to make a personalized content clip by denoting favorite segments from an original version of content and stringing those segments together. That personalized content clip may then be played by being forwarded to friends or family, or by being posted to a social media platform. Unfortunately, those often innocently intended practices stray into illegality when digital rights protected content is copied and distributed to consumers who lack a right to access that content, such as a license or other entitlement to the content. Moreover, and as also stated above, although legally playable content clips can be obtained from some sources, those clips are typically produced with promotional intent and reflect the editorial preferences of the owner or owners of the content, rather than its consumers.

(10) By contrast, the present application advantageously discloses a digital rights protected content playing solution that improves upon the conventional art by enabling a user to enjoy a personalized content clip without engaging in illicit copying or content distribution. Instead, according to the present inventive principles, a user may produce a playlist for a content clip (hereinafter “content compilation”) that is personalized and includes metadata identifying the content segments to be included in the content compilation, as well as source locators, such as Uniform Resource

Identifiers (URIs), for each of the content segments, but not those content segments themselves. The systems disclosed in the present application first confirm that the user has rights to access the identified segments, and then produces the playlist listing only those content segments to which the user has an access right. The playlist may then be output by such a system to the user producing the playlist (hereinafter “first user”), or to another user (hereinafter “second user”).

(11) The first user having produced the playlist may submit the playlist to the system at any subsequent time, resulting in substantially immediate playout of the content compilation corresponding to the playlist produced for the first user, in some implementations without further digital rights verification. If the second user receives the playlist, either from the system or from the first user, and wishes to enjoy a content compilation personalized by the first user, the second user may submit the playlist to the system. The system then determines whether the second user has access rights to the content segments identified by the playlist produced for the first user and generates a content compilation for the second user based on that playlist, but omitting those content segments to which the second user lacks an access right.

(12) In some implementations, the systems disclosed by the present application may notify the second user which content segments identified by the playlist produced for the first user the second user lacks a right to access, before generating the content compilation for the second user.

Moreover, in some implementations, such a system may offer the second user an opportunity to obtain the rights to access the content segments that the second user initially lacks.

(13) In some implementations, the systems and methods disclosed by the present application may be substantially or fully automated. It is noted that, as used in the present application, the terms “automation,” “automated,” and “automating” refer to systems and processes that do not require the participation of a human analyst or editor. Although, in some implementations, a human system administrator may review the performance of the automated systems and the automated methods described herein, that human involvement is optional. Thus, the methods described in the present application may be performed under the control of hardware processing components of the disclosed automated systems.

(14) FIG. 1 shows an exemplary system for performing digital rights protected content playing, according to one implementation. As shown in FIG. 1, system **100** includes computing platform **102** having hardware processor **104** and system memory **106** implemented as a non-transitory storage device. According to the present exemplary implementation, system memory **106** stores software code **120** providing graphical user interface (GUI) **130**, recommendation engine **112**, and playlist database **115**.

(15) As further shown in FIG. 1, system **100** is implemented within a use environment including communication network **108**, first user system **150a** including display **152a**, first user **154** of first user system **150a**, second user system **150b** including display **152b**, and second user **156** of second user system **150b**. In addition, FIG. 1 shows content source **140**, content **142a**, **142b**, and **142c** obtainable from content source **140**, advertisement source **144**, entitlement database **146** storing entitlement portfolios **148a** and **148b**, content metadata **114**, playlist **116**, and content compilations **158a** and **158b** each corresponding to playlist **116**. Also shown in FIG. 1 are network communication links **118** of communication network **108** interactively connecting system **100** with content source **140**, advertisement source **144**, entitlement database **146**, and first and second user systems **150a** and **150h**. It is noted that although in some implementations, entitlement database **146** may merely be accessible to system **100** via communication network **108** and network communication links **118**, in other implementations, entitlement database **146** may be included as part of system **100**.

(16) It is further noted that although only one content source, one advertisement source, and one entitlement database are shown in FIG. 1 in the interests of conceptual clarity, in other implementations, each of content source **140**, advertisement source **144**, and entitlement database **146** may correspond to multiple instances of respective content sources, advertisement sources, and

entitlement databases. In addition, although FIG. 1 depicts three items of content obtainable from content source **140**, once again that representation is provided merely in the interests of conceptual clarity. More generally, content source **140** may correspond to one or more content providers capable of providing hundreds, thousands, or tens of thousands of content items.

(17) Moreover, content **142a**, **142b**, and **142c** may include one or more of a variety of different types of media content. For example, **142a**, **142b**, and **142c** may take the form of one or more of movie content, photographs, podcast content, television (TV) programming content or other episodic media content, music content, audio book content, digital book content, digital news content, or video game content.

(18) It is also noted that entitlement portfolio **148a** in entitlement database **146** is an entitlement portfolio of first user **154**, identifying content to which first user **154** has an access right, while entitlement portfolio **148b** is an analogous entitlement portfolio of second user **156**. As defined for the purposes of the present application, the feature “entitlement portfolio” refers to a record of all relevant content to which an entitlement in the form of ownership and/or temporary use has been obtained by a user across one or more content sources **140**. Thus, for example, where content **142a**, **142b**, and **142c** are in the form of movies, entitlement portfolio **148a** of first user **154** identifies movies previously purchased and/or rented by first user **154** from one or more content sources **140**.

(19) Although the present application refers to software code **120** as being stored in system memory **106** for conceptual clarity, more generally, system memory **106** may take the form of any computer-readable non-transitory storage medium. The expression “computer-readable non-transitory storage medium,” as used in the present application, refers to any medium, excluding a carrier wave or other transitory signal that provides instructions to hardware processor **104** of computing platform **102**. Thus, a computer-readable non-transitory storage medium may correspond to various types of media, such as volatile media and non-volatile media, for example. Volatile media may include dynamic memory, such as dynamic random access memory (dynamic RAM), while non-volatile memory may include optical, magnetic, or electrostatic storage devices. Common forms of computer-readable non-transitory media include, for example, optical discs, RAM, programmable read-only memory (PROM), erasable PROM (EPROM), and FLASH memory.

(20) Moreover, although FIG. 1 depicts software code **120**, recommendation engine **112**, and playlist database **115** as being co-located in system memory **106**, that representation is merely provided as an aid to conceptual clarity. More generally, system **100** may include one or more computing platforms **102**, such as computer servers for example, which may be co-located, or may form an interactively linked but distributed system, such as a cloud-based system, for instance. As a result, hardware processor **104** and system memory **106** may correspond to distributed processor and memory resources within system **100**. In one such implementation, computing platform **102** may correspond to one or more web servers accessible over a packet-switched network such as the Internet, for example. Alternatively, computing platform **102** may correspond to one or more computer servers supporting a wide area network (WAN), a local area network (LAN), or included in another type of private or limited distribution network.

(21) Although first user system **150a** is shown as a smartphone and second user system **150b** is shown as a desktop computer in FIG. 1, those representations are also provided merely as examples. More generally, first and second user systems **150a** and **150b** may be any suitable mobile or stationary computing devices or systems that implement data processing capabilities sufficient to enable use of GUI **130**, support connections to communication network **108**, and implement the functionality ascribed to first and second user systems **150a** and **150b** herein. For example, in other implementations, either or both of first and second user systems **150a** and **150b** may take the form of a laptop computer, a tablet computer, a smart TV, an e-reader, or a game console, for example. Alternatively, in some implementations, either or both of first and second user systems **150a** and **150b** may take the form of a wearable personal communication device, such

as an augmented reality (AR) or virtual reality (VR) headset or glasses, a smartwatch, or another smart personal accessory.

(22) First user **154**, who may be a consumer of media content such as movie content, photographs, podcast content, television (TV) programming content or other episodic media content, music content, audio book content, digital book content, digital news content, or video game content, for example, may utilize first user system **150a** to interact with system **100** via GUI **130**. For example, first user **154** may utilize a content editing pane of GUI **130** rendered on display **152a** to generate content metadata **114** identifying content segments for use in producing playlist **116**. It is noted that display **152a** of first user system **150a**, as well as display **152b** of second user system **150b**, may take the form of a liquid crystal display (LCD), a light-emitting diode (LED) display, an organic light-emitting diode (OLED) display, or any other suitable display screen that performs a physical transformation of signals to light. It is further noted that, in some implementations, display **152a** or **152b** may be integrated with respective first and second user systems **150a** and **150b**, such as when first and second user systems **150a** and **150b** take the form of smartphones, or laptop or tablet computers for example. However, in other implementations, for example where first and second user systems **150a** and **150b** take the form of computer towers in combination with a desktop monitor, displays **152a** and **152b** may be communicatively coupled to, but not physically integrated with first and second user systems **150a** and **150b**.

(23) Hardware processor **104** of system **100** may execute software code **120** to produce playlist **116** for first user **154** based on content metadata **114** received from first user **154** and further based on entitlement portfolio **148a** of first user **154**, such that playlist **116** lists only content segments to which first user **154** has an access right. First user **154** may have such an access right to content including one or more of the content segments identified by content metadata **114** due to enrolment in a subscription service, such as Netfix® or Hulu®, for example, by virtue of having rented a license to the content from a video-on-demand (VOD) service, or by virtue of having purchased a license to the content from content source **140**.

(24) Hardware processor **104** of system **100** may further execute software code **120** to output playlist **116** to first user **154**, and/or to second user **156** identified by first user **154**. In some implementations, first user **154** may utilize playlist **116** for his/her personal entertainment. In those use cases, system **100** may generate content compilation **158a** for first user **154** using playlist **116**. Alternatively, or in addition, first user **154** may send playlist **116** with second user **156**, for example by transmitting playlist **116** to second user **156** via email, text message, or a native messaging application provided by first user system **150a**. In those use cases, second user **156** may use playlist **116** to consume content listed in playlist **116** to which second user **156** has an access right.

(25) In order to consume content compilation **158b** corresponding to playlist **116**, second user **156** may submit playlist **116** produced for first user **154** to system **100**. Hardware processor **104** may then execute software code **120** to generate content compilation **158b** based on playlist **116** and entitlement portfolio **148b** of second user **156** such that content compilation **158b** omits content segments listed in playlist **116** to which second user **156** lacks access rights. That is to say, system **100** advantageously enables first user **154** to create playlist **116** of content segments and to send playlist **116** to others, without violating the intellectual property rights of the owner or owners of the content including the content segments identified by content metadata **114**.

(26) In some implementations, and at the discretion of first user **154**, playlist **116** produced for first user **154** may be stored in playlist database **115** of system **100**, and may be made available to other users of system **100**. In one such use case, for example, second user **156** of system **100** may access playlist database **115** to enjoy content compilations corresponding to individual playlists, or to create mashups from multiple playlists produced for other users and stored in playlist database **115**.

(27) Referring now to FIG. 2, FIG. 2 shows an exemplary diagram of software code **220** suitable for use by system **100**, in FIG. 1, according to one implementation. As shown in FIG. 2, software code **220** is configured to receive content metadata **214** as an input and to provide playlist **216** as

an output via GUI **230**, as well as to receive playlist **216** as an input and to provide content compilation **258a** and/or content compilation **258b** as an output or outputs via GUI **230**. As further shown in FIG. **2**, software code **220** includes input module **222**, licensing module **224**, playlist module **226**, content compilation module **227**, output module **228**, and may include marketplace module **229**. Also shown in FIG. **2**, licensing module **224** of software code **220** is configured to communicate with entitlement database **146**, in FIG. **1**, while content compilation module **227** is configured to communicate with content source **140** and may be configured to communicate with advertisement source **144**. Moreover, when included in software code **220**, marketplace module **229** may be configured to communicate with one or more of recommendation engine **112**, playlist database **115**, and entitlement database **146**, in FIG. **1**.

(28) Content metadata **214**, playlist **216**, content compilation **258a**, and content compilation **258b**, in FIG. **2**, correspond respectively in general to content metadata **114**, playlist **116**, content compilation **158a**, and content compilation **158b**, in FIG. **1**, and those respective corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure. In addition, software code **220** and GUI **230** correspond in general to software code **120** and GUI **130**, in FIG. **1**, and those respective corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure. That is to say, like software code **220**, software code **120** may include features corresponding respectively to input module **222**, licensing module **224**, playlist module **226**, content compilation module **227**, output module **228**, and marketplace module **229**.

(29) FIG. **3** shows exemplary content editing pane **332** of GUI **330** rendered on display **352a**, according to one implementation. GUI **330** corresponds in general to GUI **130/230** in FIGS. **1** and **2**, and those features may share the characteristics attributed to any of those corresponding features by the present disclosure. Thus, although not apparent from FIGS. **1** and **2**, GUI **130/230** may be configured to provide editing pane **332** including the additional features described below. However, it is noted that those features, including the specific input fields shown in FIG. **3**, are merely exemplary. That is to say, in various implementations, content editing pane **332** of GUI **330** may include additional features and input fields not shown in FIG. **3**, or may omit one or more of the features and input fields shown in FIG. **3**. It is further noted that display **352a** on which content editing pane **332** of GUI **330** is shown to be rendered in FIG. **3** corresponds in general to display **152a** of first user device **150a**, in FIG. **1**.

(30) Referring to FIGS. **1** and **3** in combination, first user **154** may utilize content editing pane **332** of GUI **130/330** to identify content segments for inclusion in personalized content compilation **158a**. For example, first user **154** (the author of playlist **116** and corresponding content compilation **158**) is identified in Author ID field **370**. First user **154** may select content using Title field **372**, and may watch and/or listen to that content as it is played out in Playout Window **360a**. During playout of the content, first user **154** may utilize “begin segment” control **364** and “end segment” control **366** to identify a segment or segments of the content to be included in content compilation **158a**, based on timecode **362** locations of the beginning and end of the content segment, or video frame numbers, or audio track numbers, for example. Alternatively, or in addition, first user **154** may use Natural Language Search field **378** to specify one or more content segments. For example, where first user **154** desires a content compilation including action scenes drawn from all movies included in a movie franchise, first user **154** may list the movie titles or franchise name in Title field **372**, and enter “all action scenes” in Natural Language Search field **378**.

(31) In some implementations, content editing pane **332** of GUI **330** may also include one or more of Content ID field **368**, Recipient ID field **374**, and Expiration field **376**. Content ID field **368** may include a unique code or number assigned to content identified in Title field **372**. For example, in some implementations, first user may **154** enter a content title or titles in Title field **372** and Content ID field **368** may be auto-filled by software code **120/220** in FIGS. **1** and **2**, executed by hardware processor **104**.

(32) Recipient ID field **374** may be utilized by first user **154** if first user **154** wishes to have content compilation playlist **116/216** sent directly to someone else by system **100**, such as to second user **156**, for example. Otherwise, in some implementations, the default entry in Recipient ID field **374** can match the entry in Author ID field **370**, in which case content compilation playlist **116/216**, when produced by software code **120/220** for first user **154**, is sent only to first user **154**.

(33) Expiration field **376** enables first user **154** to specify how long a particular content compilation can be accessed, either by first user **154** or by second user **156** with whom first user **154** decides to send playlist **116/216**. In some implementations, the expiration of content compilation **158a/258a** or **158b/258b** corresponding to playlist **116/216** may be an absolute expiration, identified by a definite duration or by a specific date and time determined by first user **154**. However, in other implementations, the expiration of content compilation **158a/258a** or **158b/258b** corresponding to playlist **116/216** may be relative to some action. By way of example, in some implementations, the expiration of each instance of content compilation **158a/258a** or **158b/258b** may occur after an interval triggered by generation of that instance. In other words, content compilation **158a/258a** corresponding to playlist **116/216** may expire after a predetermined interval, selected by first user **154**, and triggered by generation of content compilation **158a/258a** for first user **154**. Analogously, content compilation **158b/258b** corresponding to playlist **116/216** may expire after the same predetermined interval selected by first user **154**, but triggered in that instance by generation of content compilation **158b/258b** for second user **156**.

(34) Referring to FIG. **4**, FIG. **4** will be described briefly, with a more detailed discussion of its features provided below, by reference to the flowchart shown in FIG. **5B**. FIG. **4** shows exemplary introductory message pane **434** of GUI **430**, rendered on display **452b** and including message window **480** and purchase options **482a** and **482b**, according to one implementation. GUI **430** corresponds in general to GUI **130/230/330** in FIGS. **1**, **2**, and **3**, and those features may share the characteristics attributed to any of those corresponding features by the present disclosure. Thus, although not apparent from FIGS. **1** and **2**, GUI **130/230** may be configured to provide introductory message pane **434**. However, it is noted that the specific features of introductory message pane **434** shown in FIG. **4** are merely exemplary. That is to say, in various implementations, introductory message pane **434** of GUI **430** may include additional features not shown in FIG. **4**, or may omit one or more of the features shown in FIG. **4**. It is further noted that display **452b** on which introductory message pane **434** of GUI **430** is shown to be rendered in FIG. **4** corresponds in general to display **152b** of second user device **150b**, in FIG. **1**.

(35) The functionality of software code **120/220** in FIGS. **1** and **2** will be further described by reference to FIGS. **5A** and **5B**. FIG. **5A** shows flowchart **500A** presenting an exemplary method for performance by system **100**, in FIG. **1**, according to one implementation, while FIG. **5B** shows flowchart **500B** presenting an exemplary method for performing digital rights protected content playing, according to one implementation. With respect to the methods outlined in FIGS. **5A** and **5B**, or the method outlined by the combination of FIGS. **5A** and **5B**, it is noted that certain details and features have been left out of flowcharts **500A** and **500B** in order not to obscure the discussion of the inventive aspects disclosed in the present application.

(36) Referring to FIG. **5A** in combination with FIGS. **1**, **2**, and **3** flowchart **500A** begins with receiving, by system **100**, content metadata **114/214** identifying one or more content segments from first user **154** (action **590**). The content segments identified by content metadata **114/214** may include segments of one or more of a variety of different types of media content. For example, the content segments identified by content metadata **114/214** may include photographs, or segments of one or more of movie content, podcast content, TV programming content or other episodic media content, music content, audio book content, digital book content, digital news content, or video game content.

(37) Content metadata **114/214** may be generated by first user system **150a**, in response to inputs to content editing pane **332** of GUI **130/230/330** rendered on display **152a** of first user system **150a**.

As discussed above by reference to FIG. 3, first user **154** may select content using Title field **372** of editing pane **332**, and may watch and/or listen to that content as it is played out in Playout Window **360a** of editing pane **332**. During playout of the content, first user **154** may utilize “begin segment” control **364** and “end segment” control **366** to generate metadata **114/214** identifying a desired segment or segments of the content, based on timecode **362** locations of the beginning and end of the content segment, or video frame numbers, or audio track numbers, for example. Alternatively, or in addition, first user **154** may use Natural Language Search field **378** of content editing pane **332** to specify one or more content segments. Content metadata **114/214** may be also generated in response to inputs to Recipient ID field **374** and/or Expiration field **376**, as also discussed above.

(38) As shown by FIG. 1, in one implementation, content metadata **114/214** may be received from first user system **150a** by system **100** via communication network **108** and network communication links **118**. Content metadata **114/214** may be received from first user system **150a** by input module **222** of software code **120/220**, executed by hardware processor **104**, and using content editing pane **332** of GUI **130/230/330**.

(39) Continuing to refer to FIGS. 1, 2, 3, and 5A, flowchart **500A** continues with determining whether first user **154** has a right to access at least one of the content segments identified by content metadata **114/214** (action **591**). In use cases in which first user **154** has play access to content using Playout Window **360a** of content editing pane **332**, an access right to segments of that content can be presumed. However, where content metadata **114/214** identifies content segments based on inputs by first user **154** to Natural Language Search field **378**, an access right to those content segments may or may not be held by first user **154**, and therefore cannot be presumed. Determining whether first user **154** has a right to access at least one of the content segments identified by content metadata **114/214** may be performed by licensing module **224** of software code **120/220**, executed by hardware processor **104**, and by reference to entitlement portfolio **148a** of first user **154**, stored in entitlement database **146**.

(40) Flowchart **500A** continues with producing, in response to determining that first user **154** has a right to access at least one of the content segments identified by content metadata **114/214** received in action **590**, playlist **116/216** using that content metadata, playlist **116/216** including a source locator for each of the one or more content segments to which first user **154** has an access right (action **592**). Production of playlist **116/216** using content metadata **114/214** may be performed by playlist module **226** of software code **120/220**, executed by hardware processor **104**.

(41) As noted above, a significant advantage of the systems and methods disclosed in the present application is that a system user, such as first user **154**, has complete discretion as to what segments of content to which first user **154** has an access right are to be included in personalized content compilation **158a/258a** generated for first user **154**. A further significant advantage of the present systems and methods is that, of the content segments identified by content metadata **114/214**, only those content segments to which first user **154** has an access right are listed in playlist **116/216**. That is to say, content segments identified by content metadata **114/214** to which first user **154** lacks access rights are not listed in playlist **116/216** produced for first user **154**. Consequently, the present systems and methods are able to provide first user **154** with broad artistic discretion, while concurrently protecting the intellectual property rights of the owner or owners of the content segments selected by first user **154**.

(42) In some implementations, the source locator for each content segment included in playlist **116/216** may be a URI for obtaining that content segment, such as a Uniform Resource Locator (URL) for obtaining the content segment via communication network **108** and network communication links **118**, for example. As noted above, the content segments identified by content metadata **114/214** may include photographs, or segments of one or more of movie content, podcast content, TV programming content or other episodic media content, music content, audio book content, digital book content, digital news content, or video game content. Moreover, in some implementations, those content segments included in playlist **116/216** may include a mix of

different types of content.

(43) Thus, in some implementations, playlist **116/216** may list more than one type of content segment. For example, playlist **116/216** may list one or more visual content segments, such as photographs or video, for example, and one or more audio content segments, such as music, voice over, or narration. Furthermore, in some implementations, playlist **116/216** may specify that a visual content segment and an audio content segment be played out concurrently. As a specific example, playlist **116/216** may include a segment from a movie based on a book, as well as a segment of that book in audio book form narrated by the author. In that use case, content playlist **116/216** may specify that the movie segment and audio segment be played out concurrently so that the author may seem to be describing the movie.

(44) Flowchart **500A** can conclude with outputting playlist **116/216** to one or both of first user **154** and second user **156** (action **593**). In many use cases, playlist **116/216** produced by system **100** for first user **154** is sent solely to first user **154**, who may then send playlist **116/216** to other users, such as second user **156**. However, as shown in FIG. 3, in some implementations, first user **154** may identify another recipient of playlist **116/216**, either in addition to, or in lieu of first user **154**, by identifying that recipient in Recipient ID field **374** of content editing pane **332** provided by GUI **130/230/330**.

(45) Action **593** may be performed by output module **228** of software code **120/220**, executed by hardware processor **104**. For example, and as shown in FIG. 1, playlist **116/216** may be transmitted to one or both of first user device **150a** utilized by first user **154** and second user device **150b** utilized by second user **156**, via communication network **108** and network communication links **118**.

(46) In implementations in which playlist **116/216** is output to first user **154**, first user **154** may utilize playlist **116/216** to consume personalized content compilation **158a/258a**. For example, first user **154** may submit playlist **116/216** to system **100** via GUI **130/230/330**, communication network **108**, and input module **222** of software code **120/220**. Hardware processor **104** of system **100** may then execute content compilation module **227** to generate personalized content compilation **158a/258a** for first user **154** and to playout personalized content compilation **158a/258a** using output module **228** and GUI **130/230/330**.

(47) In some use cases, first user **154** may wish to obtain content compilation **158a/258a** of favorite scenes from multiple movies, or favorite songs from multiple albums, for example. However, in other use cases, first user **154** may use system **100** to modify an existing creative work product to make the content included in that work product more enjoyable, or more suitable for a particular audience. As one specific example, first user **154** may edit a PG rated movie including some violence or scary scenes by using system **100** to edit out all scary scenes or scenes including violence to make the movie more suitable for enjoyment by children. As another example, first user **154** may edit an album of rap music by editing out lyrics that are profane or obscene.

(48) As noted above, in some use cases, enjoyment of content compilation **158a/258a** by first user **154** may include the ability to play some or all of that content with another user, such as second user **156**. In those use cases, after receiving playlist **116/216** from system **100**, first user **154** may send playlist **116/216** with second user **156** by transmitting playlist **116/216** to second user **156** via text message email, or any native messaging application on first user device **150a**, for example. Second user **156** may then use system **100** to generate content compilation **158b/258b** for second user **156** based on playlist **116/216** produced for first user **154**, as outlined by flowchart **500B** in FIG. 5B.

(49) Referring to FIG. 5B in combination with FIGS. 1 and 2, flowchart **500B** begins with receiving, by system **100** from second user **156**, playlist **116/216** produced for first user **154** (action **594**). As noted above, playlist **116/216** includes metadata identifying one or more content segments in the form of one or more of photographs, and/or segments of one or more of movie content, podcast content, TV programming content or other episodic media content, music content, audio

book content, digital book content, digital news content, or video game content.

(50) As shown by FIG. 1, in one implementation, playlist **116/216** may be received from second user system **150b** by system **100** via communication network **108** and network communication links **118**. Playlist **116/216** may be received from second user system **150b** by input module **222** of software code **120/220**, executed by hardware processor **104**.

(51) Continuing to refer to FIGS. 1, 2, and 5B, flowchart **500B** continues with determining whether second user **156** has a right to access each of the content segments identified by playlist **116/216** (action **595**). Determining whether first user **156** has a right to access each of the content segments identified by playlist **116/216** may be performed by licensing module **224** of software code **120/220**, executed by hardware processor **104**, and by reference to entitlement portfolio **148b** of second user **156**, stored in entitlement database **146**.

(52) In use cases in which second user **156** lacks a right to access one or more of the content segments identified by playlist **116/216**, the method outlined by flowchart **500B** may optionally include informing second user **156** which content segments second user **156** lacks access rights to (action **596**). Moreover, in some implementations, flowchart **500B** may further optionally include offering second user **156** the opportunity to acquire the access rights that second user **156** lacks (action **597**).

(53) For example, referring to FIG. 4, action **595** may be followed by actions **596** and **597** in which introductory message pane **434** is presented to second user **156** via GUI **130/230/430** and display **452b** of second user system **150b**. As shown in FIG. 4, message window **480** of introductory message pane **434** informs second user **156** that second user **156** lacks a right to “Movie A” and “Audio Book B” to which content segments identified by playlist **116/216** correspond.

(54) According to the exemplary implementation shown in FIG. 4, introductory message pane **434** also provides second user **156** with purchase option **482a** for the right to access “Movie A” and purchase option **482b** for the right to access “Audio Book B.” Message window **480** also informs second user **156** that any content segments identified by playlist **116/216** to which second user **156** lacks access rights will be skipped during playout of content compilation **158b/258b**. When included in the method outlined by flowchart **500B**, actions **596** and **597** may be performed by licensing module **224** and output module **228** of software code **120/220**, executed by hardware processor.

(55) Flowchart **500B** continues with generating content compilation **158b/258b** using playlist **116/216**, content compilation **158b/258b** omitting any content segment identified by playlist **116/216** to which second user **156** lacks access rights (action **598**). Generation of content compilation **158b/258b** using playlist **116/216** may be performed by content compilation module **227** of software code **120/220**, executed by hardware processor **104**. For example, and as shown by FIG. 2, content compilation module **227** may utilize the source locators included in playlist **116/216** to obtain the content segments identified by playlist **116/216** and to which second user **156** has access rights from one or more content sources **140**.

(56) It is contemplated that in some implementations, first user **154** and second user **156** may each have a subscription to the same streaming service, but those respective subscriptions may be at different service levels or impose different constraints. For example, first user **154** may have a commercial free subscription to “Streaming Service 1,” while second user **156** may have a subscription to the same streaming service that requires the playout of commercials. In such a use case, playlist **116/216** produced for first user **154**, as well as content compilation **158a/258a** generated for first user **154**, may omit advertising content. However, in that use case, content compilation **158b/258b** generated for second user **156** using playlist **116/216** produced for first user **154** may nevertheless include advertising content omitted from playlist **116/216** due to the different constraints arising from the subscription status of second user **156**. Conversely, where the subscription status of each of first user **154** and second user **156** is reversed from that described above, content compilation **158b/258b** generated for second user **156** using playlist **116/216**

produced for first user **154** may omit advertising content included in playlist **116/216**.

(57) As another exemplary use case, first user **154** may use system **100** to produce playlist **116/216** of digital news articles to track how an important news story has been reported across different news outlets and/or over time. First user **154** may wish to send that compilation of digital news with second user **156** and may email, text, or use any suitable native messaging application on first user device **150a** to send playlist **116/216** to second user **156**. System **100** may then receive playlist **116/216** produced for first user **154** from second user **156**. In that use case, action **595** may include not only determining that second user **156** has an access right to the news sources referenced by playlist **116/216**, but that exercising that access right will not exceed a predetermined quota of content items that second user **156** can access for free, or that might otherwise incur charges or penalties for second user **156**.

(58) Where a content segment listed in playlist **116/216** would cause second user **156** to **115** exceed a quota or incur a charge or penalty, system **100** may simply omit that content segment from content compilation **158b/258b** during playout of content compilation **158b/258b** to second user, thereby effectively skipping over one or more content segments identified by playlist **116/216**. Alternatively, hardware processor **104** may execute software code **120/220** to substitute interstitial content segments for those content segments identified by playlist **116/216** but omitted from content compilation **158b/258b** generated for second user **156**. Such “replacement content” may take the form of promotional content for the omitted content segment, and/or may include advertising content.

(59) Flowchart **500A** can conclude with playing out content compilation **158b/258b** to second user **156** (action **599**). Action **599** may be performed by output module **228** of software code **120/220**, executed by hardware processor **104**. For example, and as shown in FIG. **1**, content compilation **158b/258b** may be streamed or otherwise transmitted to second user device **150b** utilized by second user **156**, via communication network **108** and network communication links **118**.

(60) In some implementations, system **100** may recommend content to one or both of first user **154** and second user **156** based on playlist **116/216** produced for first user **154**. Referring to FIG. **6**, FIG. **6** shows exemplary content promotion pane **636** of GUI **630**, rendered on display **652** and including message windows **680a** and **680b**, purchase option **682**, and content compilation selection option **683**, according to one implementation. Also shown in FIG. **6** are items of content **642a**, **642b**, and **642c**, as well as previously produced playlists **616a**, **616b**, and **616c**.

(61) GUI **630** corresponds in general to GUI **130/230/330/430** in FIGS. **1**, **2**, **3**, and **4**, and those features may share the characteristics attributed to any of those corresponding features by the present disclosure. Thus, although not apparent from FIGS. **1** and **2**, GUI **130/230** may be configured to provide content promotion pane **636**. However, it is noted that the specific features of content promotion pane **636** shown in FIG. **6** are merely exemplary. That is to say, in various implementations, content promotion pane **636** of GUI **630** may include additional features not shown in FIG. **6**, or may omit one or more of the features shown in FIG. **6**. It is further noted that display **652** on which content promotion pane **636** of GUI **630** is shown to be rendered in FIG. **6** corresponds in general to either or both of display **152a** of first user device **150a** and display **152b** of second user device **150b**, in FIG. **1**.

(62) As shown in FIG. **6**, message window **680a** of content promotion pane **636** informs a user of system **100**, i.e., first user **154** or second user **156**, of content **642a**, **642b**, and **642c** recommended as being likely of interest to the user based on playlist **116/216**, and to which the user presently lacks access rights. As further shown in FIG. **6**, message window **680b** of content promotion pane **636** informs the user of content compilations **616a**, **616b**, and **616c** previously produced for other users, made available for playing by those users, and recommended as being likely of interest to the user based on playlist **116/216**.

(63) It is noted that content **642a**, **642b**, and **642c** correspond respectively in general to content **142a**, **142b**, and **142c**, in FIG. **1**, and those respective corresponding features may comprise any of

the characteristics attributed to either corresponding feature by the present disclosure. Thus, like content **142a**, **142b**, and **142c**, content **642a**, **642b**, and **642c** may take the form of one or more of movie content, photographs, podcast content, TV programming content or other episodic media content, music content, audio book content, digital book content, digital news content, or video game content.

(64) According to the exemplary implementation shown in FIG. 6, content promotion pane **636** provides the user with purchase option **682** for the right to acquire an entitlement to one or more of content **642a**, **642b**, and **642c**. In addition, content promotion pane **636** provides the user with playlist selection option **683** enabling the user to select one or more of playlists **616a**, **616b**, and **616c**. For example, according to one use case, the user can create a content compilation mashup using a combination of playlists selected from playlists **616a**, **616b**, and **616c**.

(65) Content promotion pane **636** of GUI **630** may be provided by marketplace module **229** of software code **120/220**, executed by hardware processor. For instance, and as shown by FIG. 2, marketplace module **229** may utilize recommendation engine **112** and playlist database **115** to identify playlists **616a**, **616b**, and **616c** likely to be of interest to the user, and may utilize recommendation engine **112** and entitlement database **146** to identify content **642a**, **642b**, and **642c** likely to be of interest to the user and to which the user presently lacks access rights.

(66) Thus, the present application discloses systems and methods for performing digital rights protected content playing that confer significant improvements over the conventional art. According to the inventive concepts disclosed in the present application, a first user may produce a playlist for a personalized content compilation that includes metadata identifying the content segments to be included in the content compilation, as well as source locators for each of the content segments, but not those content segments themselves. The systems and methods disclosed in the present application first confirm that the first user has rights to access the identified segments, and then produces the playlist so as to include only metadata for those content segments to which the first user has an access right.

(67) If a second user wishes to enjoy the content compilation personalized by the first user, the second user may submit the playlist to the system. According to the systems and methods disclosed herein, the system then determines whether the second user has rights to access the content segments identified by the playlist produced for the first user, and generates a content compilation for the second user based on that playlist, but omitting those content segments to which the second user lacks an access right. Consequently, the systems and methods disclosed in the present application advantageously enable users to create playlists of content segments and to send those playlists to other users, without violating the intellectual property rights of the owner or owners of the content from which the content segments are extracted.

(68) From the above description it is manifest that various techniques can be used for implementing the concepts described in the present application without departing from the scope of those concepts. Moreover, while the concepts have been described with specific reference to certain implementations, a person of ordinary skill in the art would recognize that changes can be made in form and detail without departing from the scope of those concepts. As such, the described implementations are to be considered in all respects as illustrative and not restrictive. It should also be understood that the present application is not limited to the particular implementations described herein, but many rearrangements, modifications, and substitutions are possible without departing from the scope of the present disclosure.

Claims

1. A system comprising: a computing platform having a hardware processor and a system memory storing a software code; the hardware processor configured to execute the software code to: receive, from a first user, metadata identifying a plurality of contents; produce a first content

compilation for the first user using the metadata, the first content compilation identifying the plurality of contents; output the first content compilation to the first user; receive, from a second user, information identifying the first content compilation produced for the first user; determine one or more of the plurality of contents identified by the first content compilation that the second user lacks a right to access; generate a second content compilation for the second user using the first content compilation, the second content compilation omitting the one or more of the plurality of contents identified by the first content compilation that are determined the second user lacks the right to access; and play out the second content compilation to the second user.

2. The system of claim 1, wherein the plurality of contents comprise at least one visual content and at least one audio content.
3. The system of claim 2, wherein the first content compilation specifies that the at least one visual content and the at least one audio content be played out concurrently.
4. The system of claim 1, wherein the plurality of contents comprise at least one of movie content, photographs, podcast content, television programming content, music content, audio book content, digital book content, digital news content, or video game content.
5. The system of claim 1, wherein the first content compilation includes a source locator for the at least one of the plurality of contents.
6. The system of claim 1, wherein the hardware processor is further configured to execute the software code to: inform the second user, before generating the second content compilation, of each of the determined at least one of the plurality of contents to which the second user lacks the right to access; and offer the second user, before generating the second content compilation, an opportunity to acquire the right to access.
7. The system of claim 1, wherein the second content compilation includes advertising content omitted from the first content compilation.
8. The system of claim 1, wherein the second content compilation includes replacement content for each of the determined at least one of the plurality of contents to which the second user lacks the right to access.
9. The system of claim 8, wherein the replacement content is one of promotional content for advertising content or each of the determined at least one of the plurality of contents.
10. A method comprising: receiving, from a first user, metadata identifying a plurality of contents; producing a first content compilation for the first user using the metadata, the first content compilation identifying the plurality of contents; outputting the first content compilation to the first user; receiving, from a second user, information identifying the first content compilation produced for the first user; determining one or more of the plurality of contents identified by the first content compilation that the second user lacks a right to access; generating a second content compilation for the second user using the first content compilation, the second content compilation omitting the one or more of the plurality of contents identified by the first content compilation that are determined the second user lacks the right to access; and playing out the second content compilation to the second user.
11. The method of claim 10, wherein the plurality of contents comprise at least one visual content and at least one audio content.
12. The method of claim 11, wherein the first content compilation specifies that the at least one visual content and the at least one audio content be played out concurrently.
13. The method of claim 10, wherein the plurality of contents comprise at least one of movie content, photographs, podcast content, television programming content, music content, audio book content, digital book content, digital news content, or video game content.
14. The method of claim 10, wherein the first content compilation includes a source locator for the at least one of the plurality of contents.
15. The method of claim 10, further comprising: informing the second user, before generating the second content compilation, of each of the determined at least one of the plurality of contents to

which the second user lacks the right to access; and offering the second user, before generating the second content compilation, an opportunity to acquire the right to access.

16. The method of claim 10, wherein the second content compilation includes advertising content omitted from the first content compilation.

17. The method of claim 10, wherein the second content compilation includes replacement content for each of the determined at least one of the plurality of contents to which the second user lacks the right to access.

18. The method of claim 17, wherein the replacement content is one of promotional content for advertising content or each of the determined at least one of the plurality of contents.

19. A system comprising: a computing platform having a hardware processor and a system memory storing a software code; the hardware processor configured to execute the software code to: receive, from a first user, metadata identifying a plurality of contents; produce a first content compilation for the first user using the metadata, the first content compilation identifying the plurality of contents; output the first content compilation to the first user; receive, from a second user, information identifying the first content compilation produced for the first user; determine one or more of the plurality of contents identified by the first content compilation that the second user lacks a right to access; generate a second content compilation for the second user using the first content compilation, the second content compilation including advertising content omitted from the first content compilation when the second content compilation also includes a first one of the one or more of the plurality of contents identified by the first content compilation that are determined the second user lacks the right to access; and playout the second content compilation to the second user.

20. The system of claim 19, wherein the hardware processor is further configured to execute the software code to: omit, from the second content compilation, a second one of the one or more of the plurality of contents identified by the first content compilation that are determined the second user lacks the right to access.
